SYED SHAYAN ALI SHAH

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EDUCATION

2018–2020 **Master of Science in Computer Science**
Institute of Management Sciences, Peshawar, Pakistan

Academic Performance: CGPA: 3.90/4.00; Rank: 1st in a cohort of 60 students; Gold Medalist

2011–2016 **Bachelor of Science in Computer Science**
University of Peshawar, Peshawar, Pakistan

Academic Performance: CGPA: 3.60/4.00; Rank: 3rd in a cohort of 120 students.

PROFESSIONAL EXPERIENCE

Nov. 2018–Present **Lecturer in Computer Science**
Government College of Management Sciences, Peshawar, Pakistan

Responsibilities: The role involves undergraduate teaching, final-year project supervision, curriculum development, academic administration, and institutional information systems. My responsibilities have included:

- Teaching undergraduate computer science courses and designing lectures, laboratory activities, assessments, and project-based learning tasks.
- Supervising approximately three undergraduate final-year projects each academic year, including projects in computer vision, deep learning, and artificial intelligence, with guidance on methodology, implementation, experimentation, and technical reporting.
- Serving as **Focal Person for the Directorate of Commerce and Management Sciences Education** for coordination and implementation of the Online Admission System (admission.hed.gkp.pk).
- Contributing to the digital transformation of approximately **78,000 Intermediate and undergraduate admission applications**, replacing traditional paper-based processing with a centralized online system.
- Supporting process automation that reduced average application submission time from approximately **15 minutes to 1 minute** and shortened the overall admission cycle from approximately **30–40 days to 10 days**.
- Serving as a curriculum developer and DACUM curriculum expert, including development of the *Introduction to Programming in Python* and *Introduction to Database Systems* curricula for the Khyber Pakhtunkhwa Board of Technical Education.

RESEARCH PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Shah, Syed Shayan Ali**, Saeed, F., Raza, M. U., Rehman, A., Shaheryar, M., Kim, I.-M., Yun, S., and Kang, J.-M. "Plant-CLR: Contrastive Self-Supervised Pretraining for Generalizable Plant Disease Detection." *Scientific Reports*,2026. doi:10.1038/s41598-026-45684-x.

2. Malik, A., Mian Qaisar, S., **Shah, Syed Shayan Ali**, Uddin, M., and Salhi, A. "PPTM-IoV: A Privacy-Preserving Trust Management Framework for the Internet of Vehicles (IoV)." *Scientific Reports*, 2026. doi:10.1038/s41598-026-59399-6.

Peer-Reviewed Conference Papers

1. Saeed, F., Park, D., **Shah, Syed Shayan Ali**, Lee, J., Rehman, A., Kang, J.-M., Kim, I.-M., Yun, S., and Moon, S. "Secure and Privacy-Preserving Intrusion Detection in Wireless Sensor Networks via Perturbation Adversarial Training." In *Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing*, ACM, 2026, pp. 2081–2087. doi:10.1145/3748522.3779960.

2. **Shah, Syed Shayan Ali**, Anwar, H., Alnuaimi, M. A. S., Alshehhi, M. M. A., Isa, M., Al Harthi, O. J. S. A., and Ullah, F. "Beyond Labels: Contrastive Representation Learning for Scalable Image-Based Plant Disease Diagnosis." In *2025 International Congress on Smart Agriculture and Sustainable Systems (SmartAgri & SuSY)*, IEEE, 2025. doi:10.1109/SmartAgriSuSY68475.2025.11466957.

Manuscripts Under Review

1. **Shah, Syed Shayan Ali**, Khalid, A., Farman, H., "A Convolutional–Transformer Integrated Framework for Diabetic Retinopathy Screening from Fundus Images." Submitted to *Scientific Reports*, 2026. *Minor Revision*.

2. **Shah, Syed Shayan Ali**, Ali, Alahmad, R., Karshiev, S., and Saeed, F. "CASR-Net: Confidence-Aware Scribble Refinement for Weakly Supervised Lung Lesion Segmentation in Chest CT." Submitted to the *IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2026. *Under Review*.

Manuscripts in Preparation

1. **Shah, Syed Shayan Ali**, et al. "CASR-Net: Confidence-Aware Scribble Refinement for Weakly Supervised Lung Lesion Segmentation in Chest CT (Extended Journal Version)." *Manuscript in Preparation*, 2026.
   Target Journal: *IEEE Transactions on Medical Imaging (IEEE TMI)*.
   Research Focus: Confidence-aware pseudo-label refinement, teacher–student learning, uncertainty-aware supervision, and weakly supervised medical image segmentation.
   Repository: CASR-Net GitHub Repository.

2. **Shah, Syed Shayan Ali**. "EviCT: Evidence-Calibrated Interactive CT." *Working Title; Manuscript in Preparation*, 2026.
   Target Journal: *IEEE Transactions on Biomedical Engineering (IEEE TBME)*.
   Research Focus: Evidence-calibrated interactive medical image analysis, uncertainty-aware segmentation, explainable artificial intelligence, and evidence-grounded textual reasoning.
   Repository: EviCT GitHub Repository.

TEACHING EXPERIENCE

Spring 2026	*Object-Oriented Programming; Human-Computer Interaction; Introduction to Machine Learning*
Fall 2025	*Data Structures; Advanced Database Systems*
Spring 2025	*Introduction to Programming (Python); Introduction to Database Systems*

SKILLS AND COMPETENCIES

- Deep learning for image classification and semantic segmentation
- Convolutional neural networks and Vision Transformer architectures
- Self-supervised and contrastive representation learning
- Computer vision for image analysis, localization, and feature extraction
- Weakly supervised learning and segmentation
- Explainable artificial intelligence using Grad-CAM and attention visualization
- PyTorch, PyTorch Lightning, TorchVision, timm, and Hugging Face Transformers
- Python, C++, C#, SQL Server, and PostgreSQL
- Scientific data analysis using NumPy, pandas, and scikit-learn
- Research methodology, scientific writing, and technical documentation

LEADERSHIP EXPERIENCE

2014–2016 **President, Bitwise Society**
Department of Computer Science, University of Peshawar, Pakistan
Led the student computing society; organized programming workshops and coding sessions; mentored junior students; and coordinated academic and technical activities.

AWARDS AND HONORS

- **Gold Medal and First Position**, MS Computer Science, Institute of Management Sciences, Peshawar, 2020.
- **Third Position**, BS Computer Science, University of Peshawar; ranked 3rd in a cohort of 120 students, 2016.

LANGUAGE PROFICIENCY

English	IELTS Academic, overall band score **7.0** (Aug, 2025)
Urdu	Native proficiency
Pashto	Native proficiency

REFERENCES

Dr. Afsheen Khalid
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Peshawar, Pakistan
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